Web Development

HTML

❖ HTML Color Coding (use CSS)

- In HTML, you can specify colors using various methods, including color names, hexadecimal color codes, RGB values, and HSL values. Here's an overview of these methods:-

Text Color only

Color Names:

- Syntax→ This text is red.

Hexadecimal color codes:

- Syntax → This text is red.

RGB color values:

- Syntax → This text is red.

Background color only

- Syntax → BG Color name = Blue

Background color and text color mix

- Syntax →   Color name = white

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Heading check</title>
    </head>
    <body>
        <h1 style="color: red;"> Color name = Red </h1>
        <p style="color: #0000ff">Color name = Blue</p>
        <div style="color:  rgb(0,255,0);"> Color name = Green</div>
         <p style="background-color: #0000ff">BG Color name = Blue</p>
         <div style="background-color: #0000ff"> <p style="color: White;"> Color name = white</p></div>
         <p style="background-color: #ccdd35">BG Color name = yellow</p>
         <p style="background-color: red">BG Color name = red</p>


     </body>
</html>
```

Color name = Red

Color name = Blue

Color name = Green

BG Color name = Blue

Color name = white

BG Color name = yellow

BG Color name = red

❖ Concept of lists

Three types of lists

1. Unordered lists (<ul>)

- Unordered lists are used to represent lists of items that don't have a specific numerical or sequential order.
- The <ul> element is used to define an unordered list, and each list item is enclose within <li> (list item) tags.
- Unordered lists,which have no inherent (fixed or natural) order and each item is bydefault bulleted/Disk(•) and you can also use rectangle, circle or none form.

- Syntax →

```
<ul>
    <li> Roll no 1 </li>
    <li> Roll no 2 </li>
    <li> Roll no 3 </li>
</ul>
```

```
<ul type="square">

    <li> Roll no 1 </li>
    <li> Roll no 2 </li>
    <li> Roll no 3 </li>
</ul>
```

```
<ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JS</li>
</ul>
<ul type="square">
    <li>HTML</li>
    <li>CSS</li>
    <li>JS</li>
</ul>
```

- HTML
- CSS
- JS

- HTML
- CSS
- JS

2. Ordered lists (<ol>)

- Ordered lists are used for enumerating items in a specific numerical or sequential manner.
- Each item in an ordered list is usually prefaced by numbers or letters that signify their position.
- Ordered lists, which have an inherent order and each item is numbered.
- In type attribute use this → <ol type="1"> list </ol>
  1 → Numbers (Default)
  A → Capital Letters (A, B, C)
  a → Small Letters (a, b, c)
  I → Capital Roman (I, II, III)
  i → Small Roman (i, ii, iii)
- Syntax →

```
<ol >
    <li> HTML </li>
    <li> CSS </li>
    <li> JS </li>
</ol>
```

```
<ol type="i">
    <li> HTML </li>
    <li> CSS </li>
    <li> JS </li>
</ol>
```

```
<ol type="A">
    <li> HTML </li>
    <li> CSS </li>
    <li> JS </li>
</ol>
```

1. HTML
2. CSS
3. JS

i. HTML
ii. CSS
iii. JS

A. HTML
B. CSS
C. JS

➢ Nexted list

```
<ol type="i">
    <li> HTML </li>
        <ul>
            <li> HTML day1 </li>
            <li> HTML day2</li>
            <li>HTML day3</li>
        </ul>
    <li> CSS </li>
    <li> JS </li>
</ol>
```

i. HTML
- HTML day1
- HTML day2
- HTML day3

ii. CSS
iii. JS

3. Definition lists (<dl> <dt> <dd>)

- Definition lists are used to represent a list of terms (definitions) and their corresponding descriptions or definitions.
- <dl> use for definition lists, <dt> use for definition term & <dd> use for definition description
- Syntax →

```
<dl>
    <dt> HTML</dt>
        <dd> Hypertext Markup Language (HTML) is the standard markup language for documents designed to be displayed in a web browser. It defines the content and structure of web
```

content. It is often assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript.
</dd>
<dt> CSS </dt>
<dd> Cascading style sheets </dd>
<dt>JavaScript</dt>
<dd> A programming language for the web</dd>
</dl>

```html
<html lang="en">
    <body>
        <h1> Unordered list </h1>
        <ul>
            <li>Roll No 1</li>
            <li>Roll No 2</li>
            <li>Roll No 3</li>
        </ul>
        <h1> Ordered list </h1>
        <ol>
            <li>Roll No 1</li>
            <li>Roll No 2</li>
            <li>Roll No 3</li>
        </ol>
        <h1> Difinition list </h1>
        <dl>
            <dt> HTML</dt>
            <dd>Hypertext Markup Language</dd>
            <dt> CSS </dt>
            <dd> Cascading style sheets </dd>
            <dt>JavaScript</dt>
            <dd> A programming language for the web</dd>
        </dl>
    </body>
</html>
```

Unordered list

- Roll No 1
- Roll No 2
- Roll No 3

Ordered list

1. Roll No 1
2. Roll No 2
3. Roll No 3

Difinition list

HTML
Hypertext Markup Language
CSS
Cascading style sheets
JavaScript
A programming language for the web

❖ HTML Hyperlink

- In HTML, you can create hyperlinks (or links) to navigate between web pages, resources, or locations on the internet.
- Hyperlinks are created using the (anchor) element.
- <a> → “a” means anchor. The <a> element is used to define the hyperlink.
- href → “href” means hyper reference. The href attribute specifies the URL or destination of the link. It can be an absolute URL (e.g., "https://www.example.com") or a relative URL/local files link (e.g., "page.html").

- Syntax → <a href =”URL”> link text</a>

Link text

- Indeed, the link text is the visible text displayed on a web page, and it acts as the clickable link.

Example of hyperlink in HTML

- **Linking to an external website**

  Open same tab in browser

✓ <a href ="https://hira24y7.github.io/bcanotes/"> Visit BCA Notes (online mode)</a> (open same tab in browser)

  open other tab in browser

✓ <a href =”https://hira24y7.github.io/bcanotes/” target=”_main”> Visit BCA Notes (online mode)</a>

- **Linking to an internal page (Relative URL)**

  Same folder

<a href ="bca.html"> Bca notes (offline Mode)</a>

Other folder

<a href ="xyz/hira.jpg"> hira</a> (here xyz is a folder name, जो index.html के अंदर एक नया folder बना हुआ है)

- **Linking to an Email Address**

<a href="mailto:3in1store24y7@gmail.com"> Contact us </a>

- **Linking to an phone number**

<a href="tel:+918210150576">Call us</a>

- **Linking to a file (.pdf,.jpg,….)**

<a href="bcasyllabus.pdf"> Download PDF</a>

```
index.html > html > body > br
<!DOCTYPE html>
<html>
<body>
<h2>Hyperlink Example</h2>
<a href ="https://hira24y7.github.io/bcanotes/"> Visit BCA Notes (online mode)</a>
<a href ="bca.html"> Bca notes (offline Mode)</a><br>
<a href="3in1store24y7@gmail.com"> Contact us</a><br>
<a href="tel:+918210150576">Call us</a><br>
<a href="bcasyllabus.pdf"> Download PDF</a><br>
</body>
</html>
```

Hyperlink Example

Visit BCA Notes (online mode)
Bca notes (offline Mode)
Contact us
Call us
Download PDF

❖ Single page link in HTML page

- A Single Page Link is a link that jumps to another section of the same HTML page using #id.
- Syntax –

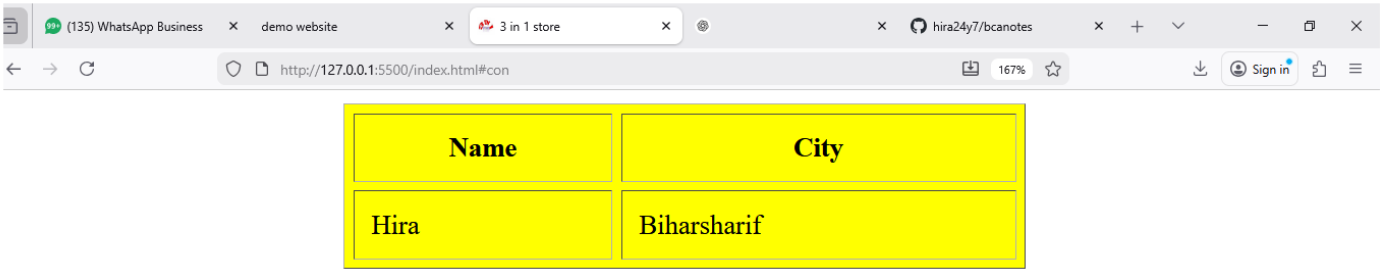
  <a href =”#id”>
  <h2 id=”123”> any text </h2>

Example → https://hira24y7.github.io/bcanotes/ (see my website)

❖ Logo link in browser tab

- Syntax- <link rel=”icon” href=”path”> (rel = relation)
- Example –

```html
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>3 in 1 store</title>
    <link rel="icon" href="3logo.jpg">
</head>
```



❖ Image and Image Mapping

1. <img> tag

- The <img> tag serves the purpose of embedding images directly within a web page.
- It is a self-contained tag, so there's no need for a closing </img> tag.
- The "src" (source) attribute is utilized to designate the source URL of the image file.
- The "alt" (alternative) attribute is responsible for supplying alternative text, in cases where the image cannot be loaded or for accessibility reasons.
- Attributes such as width and height to manage the image's dimensions. Nevertheless, it is often recommended to use CSS for styling purposes.
- Online image is called→ External Image / Absolute URL
- Local Image is called → Internal Image / Relative Path
- Syntax →

  (For Local photo / my computer photo →)

  <img src="bihar.jpg" alt="Bihar image" width="300" height ="200">

OR

<img src="bihar.jpg" alt="Bihar image" width="300" height ="200" title =”Bihar map”> (title use करने से यह हुआ की यदि कोई user फोटो पर cursor लेकर जायेगा तो जो आपने title में लिखा हैं वो दिख जायेगा)

OR

<img src="xyz/hira.jpg" height ="200">
 (here xyz is a folder name, जो index.html के अंदर एक नया folder बना हुआ है)

(For Online photo)

<img src="https://m.media-amazon.com/images/I/51gTefRkGjL._AC_UF894,1000_QL80_.jpg" alt="maa saraswati"width="250"height= "200">

Clickable image

<a href="https://hira24y7.github.io/bcanotes/"> <img src="bihar.jpg"> </a>

OR (with target)

<a href="https://hira24y7.github.io/bcanotes/" target=”_main"> <img src="bihar.jpg"> </a>

(target मतलब लिंक किस जगह open होगा -

_self → Same tab में open होगा (Default)

_blank → New tab में open होगा

_parent → Parent frame में open होगा

_top → Full window में open होगा

_main → जब page में <iframe name="main"> हो।

index.html > html > head > title

```html
<!DOCTYPE html>
<html>
    <head>
        <title> Image mapping </title>
    </head>
<body>
<a href="https://hira24y7.github.io/bcanotes/" target="_main">
    <img src="bihar.jpg" height="300"> </a>
</body>
</html>
```

2. <figcaption>

- <figcaption> tag is used to give a caption (title or description) to an image, diagram, chart, or any content inside <figure> tag.
- Syntax → <figure>
  <img> any image

<figcaption> name of image </figcaption>
</figure>

```
<figure>
    <img src="bihar.jpg" height="300">
    <figcaption>Bihar ka photo</figcaption>
</figure>
```

Bihar ka photo

✓ यदि image को center में रखना चाहते हैं तो ऐसे करें -
 <figure> any image </figure>

✓ (नोट:- HTML5 में इसे avoid करें और आप CSS का use करें)

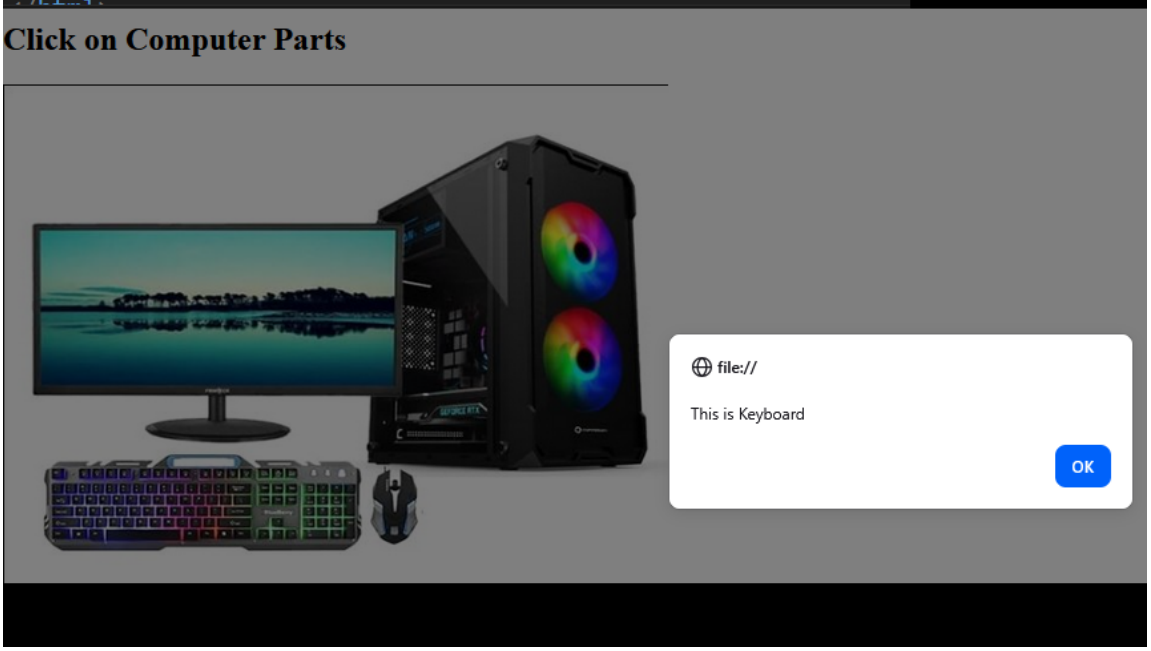
❖ Image Mapping (Image maps)

- Image mapping allows you to define clickable regions on an image, each associated with a specific action or URL.
- This is useful for creating interactive diagrams, navigation menus, or image-based forms.
- The <map> element has a name attribute that should match the usemap attribute of the associated <img>element.
- Each<area> element can have different attributes, depending on the shape of the clickable area:
    - ✓ For rectangular areas: shape, coords, and href.
    - ✓ For circular areas: shape, coords, and href.
    - ✓ For polygonal areas: shape, coords, and href.
- Syntax →

<img src="bihar.jpg" usemap="#workmap" width="400">
<map name="workmap">

<area shape="rect" coords="34,44,270,350" href="https://www.google.com">

```html
<html>
<body>
<img src="computer.jpg" usemap="#computerparts" width="500">
<map name="computerparts">
  <!-- Monitor Area -->
  <area shape="rect" coords="100,50,400,250"
        href="#"
        onclick="alert('This is Monitor')">
  <!-- CPU Area -->
  <area shape="rect" coords="420,100,480,300"
        href="#"
        onclick="alert('This is CPU Cabinet')">
  <!-- Keyboard Area -->
  <area shape="rect" coords="150,300,400,350"
        href="#"
        onclick="alert('This is Keyboard')">
  <!-- Mouse Area -->
  <area shape="rect" coords="410,320,460,370"
        href="#"
        onclick="alert('This is Mouse')">
</map>
</body>
```



<iframe> tag

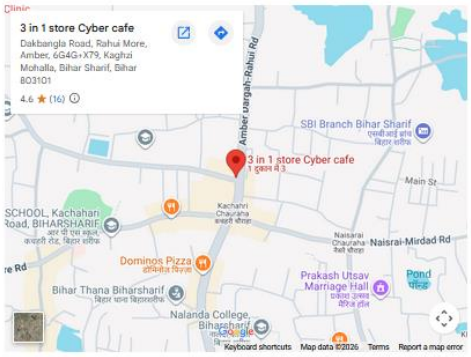
- The <iframe> (Inline Frame) tag is used to embed another web page inside the current HTML page.

- It displays a webpage, video, map, or other external content inside a frame.
- Website inside website
- Syntax → <iframe src="link"></iframe>

```
<iframe src="https://ppup.ac.in/" width="500" height="300"></iframe>
```

```
<body>
    <h1> my shop location </h1>
    <center>
    <iframe src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3609.8761095182026!2d85.5257494!3d25.207400199999995!2m3!1f0!2f0!3f0!3m2!1i1024
    width="600" height="450" style="border:0;" allowfullscreen="" loading="lazy" referrerpolicy="no-referrer-when-downgrade"></iframe>
  </center>
```

my shop location



➢ <Video> tag

- The <video> tag in HTML is used to embed video content in a web page.

➢ <source> Tag

- Used inside <video> to specify multiple video files.

➢ <track> Tag

- Used to add subtitles, captions, or descriptions.

❖ Important attributes

✓ Src → Specifies the video file
✓ Controls → Shows play, pause, volume buttons
✓ Autoplay → Video plays automatically
✓ Loop → Video repeats again and again
✓ Muted → Video starts without sound
✓ Width → Sets video width
✓ Height → Sets video hight
✓ Poster → Shows image before video plays
✓ Preload → Controls loading behavior (auto, metadata, none)

Example –

```
<body>
   <h2> Kashi Video </h2>
   <video controls height="300" muted loop="infinite"
poster="bihar.jpg">
<track src="subtitles.vtt" kind="subtitles" srclang="en" label="English">
      <source src="kashi.mp4">
   </video>
</body>
```

```html
<body>
    <h2> Kashi Video </h2>
    <video controls height="300" muted loop="infinite" poster="bihar.jpg">
        <source src="kashi.mp4">
    </video>
</body>
```

Kashi Video

➢ <audio> tag

- The <audio> tag in HTML5 is used to embed sound or music in a web page.
- Example –

```
<audio controls loop muted autoplay>
        <source src="shiv.mp3" type="audio/mp3">
</audio>
```

If provide download link then –

```
<a href="music.mp3" download>Download MP3</a>
```

```html
<body>
    <h2> Kashi Video </h2>
    <video controls height="300" muted loop="infinite" poster="bihar.jpg">
        <track src="subtitles.vtt" kind="subtitles" srclang="en" label="English">
        <source src="kashi.mp4">
    </video> <br>
    <h2> Shiv Song </h2>
    <audio controls loop muted autoplay>
        <source src="shiv.mp3" type="audio/mp3">
    </audio>
</body>
```

Kashi Video

Shiv Song

0:00 / 0:20

❖ <marquee> tag

- The <marquee> tag in HTML is used to create scrolling text or images on a web page (left, right, up, or down).
- Note :- <marquee> is deprecated (not recommended in HTML5). It is better to use CSS animations instead.
- Syntax → <marquee>Scrolling Text</marquee>

➢ Some Attributes

Attribute	Description	Example
direction	Sets scrolling direction (left(by default), right, up, down)	<marquee direction="right">Text</marquee>

behavior	Sets movement type (scroll(by default), slide, alternate)	<marquee behavior="alternate">Text</marquee>
scrollamount	Sets speed of scrolling	<marquee scrollamount="10">Text</marquee>
scrolldelay	Sets delay between movements (in milliseconds)	<marquee scrolldelay="100">Text</marquee>
loop	Number of times marquee will repeat	<marquee loop="3">Text</marquee>
bgcolor	Background color	<marquee bgcolor="yellow">Text</marquee>
width	Width of marquee	<marquee width="300">Text</marquee>
height	Height of marquee	<marquee height="50">Text</marquee>
hspace	Horizontal space around marquee i.e. align	<marquee hspace="20">Text</marquee>
vspace	Vertical space around marquee i.e. align	<marquee vspace="20">Text</marquee>

Example –

```
<marquee> Hello Hira </marquee>
   <br><br>
  <a href="BCA New Syllabus.pdf"> <marquee><font color = "blue">
Download BCA Syllabus </font>  </marquee></a>
  <br><br>
  <marquee>
   <img src="ant.jpg">
  </marquee>
```

```html
<body>
    <br><br>
    <marquee> Hello Hira </marquee>
    <br><br>
   <a href="BCA New Syllabus.pdf"> <marquee><font color = "blue"> Download BCA Syllabus </font>  </
   marquee></a>
   <br><br>
   <marquee>
    <img src="ant.jpg">
   </marquee>
```

Hello Hira

Download BCA Syllabus